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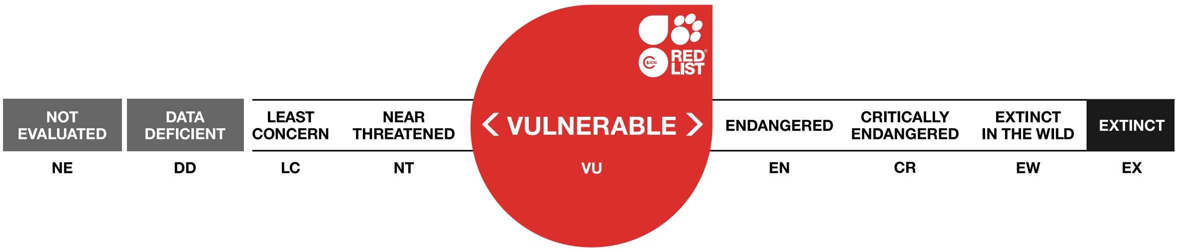
Lysurus fossatii

Assessment by: Hernandez Caffot, M.L., Niveiro, N., Pelissero, D., Maubet, Y., Sánchez, R., Ranieri, C., Kuhar, F. & Torres, D.



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Fungi	Basidiomycota	Agaricomycetes	Phallales	Phallaceae

Scientific Name: *Lysurus fossatii* Nouhra, Hern. Caff. & L.S. Domínguez

Taxonomic Source(s):

Index Fungorum Partnership. 2023. Index Fungorum. Available at: http://www.indexfungorum.org.

Assessment Information

Red List Category & Criteria: Vulnerable B2ab(iii,v) ver 3.1

Year Published: 2023

Date Assessed: May 31, 2023

Justification:

Lysurus fossatii was first recorded in 2013 in Argentina and there are only two sites in the country, Espinal forests in Córdoba province and Pampean mountains in Buenos Aires province, being considered two subpopulations. It is represented by seven collections, and the only material from Buenos Aires has been lost. It is a large species and very distinctive, and several mycological surveys have been made to the Espinal forests and to Sierra de la Ventana since its first collection, but there are no new records of the species. Also, based on known occurrences and expeditions carried in adjacent ecosystems with great sampling effort, it is thought to be likely that the species is restricted to those two locations, being considered a rare species with no clear habitat requirements. Within its range it could be restricted to an area of occupancy (AOO) of 600-800 km^2, and even if it does occur more widely in the region its AOO is not expected to exceed 2,000 km^2.

The Espinal Phytogeographic Province has been under human pressure including fires and deforestation for forestry, crop and cattle exploitation and urbanization (Bertonatti and Coruera 2000; Lewis *et al*. 2004, 2009; Zak 2008; Noy-Meir *et al.* 2012). Only 10% remains from the original vast extension of this dry environment with 1/3 of the area under strict protection (Bastin *et al*. 2017, Garachana *et al*. 2018, Zeballos *et al*. 2020). Sierra de la Ventana is considered as a unique environment with high number of endemic species severely endangered by the high rates of fire events, exotic shrub and arboreal vegetation introduction, and in the last decade the agroecosystem advance over this ecosystem has led to natural land cover change and biodiversity loss (Rodriguez Souilla and Azaro 2021, Michalijos *et al*. 2022). The population size was assumed to be no more than 10,000 mature individuals, possibly being even smaller than 2,500. Espinal forests and Sierra de la Ventana ecosystem have been devastated by land use change by fire, agroecosystem growth for forestry, crop and cattle exploitation, and urbanization. Assuming a natural land cover degradation at rates of 20% in a lapse of 20 years, it is inferred that the population size is in continuing decline. The fungal species is precautionarily assessed as Vulnerable under criterion B2ab(iii,v).

Geographic Range

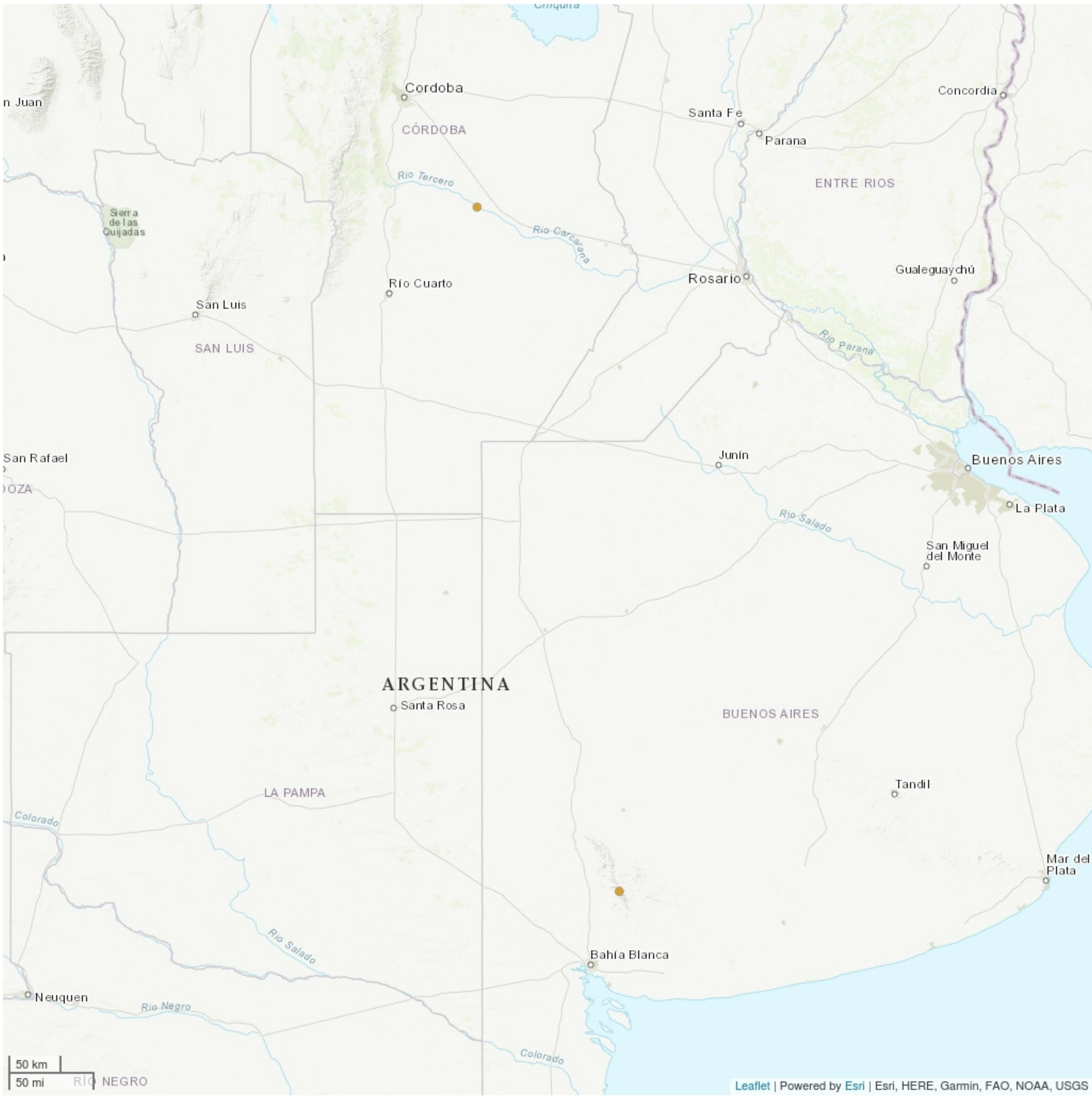
Range Description:

Lysurus fossatii has been recorded from two distinct areas. There are collections from in and around at Estancia “El Yucat”, 15 km from Tío Pujio, Departamento San Martín, central-eastern Córdoba province, central Argentina, (32° 21' 57"S; 63° 26' 09" W), where it is widespread in the area in wet and mulch-rich soil, in shady areas within shrubs and trees; and in Buenos Aires, Partido de Tronquist, at Sierra de la Ventana 38°03'45.9"S; 62°04'00.1" W), in areas directly exposed to the sun. Based on known occurrences and expeditions carried in adjacent ecosystems with great sampling effort, it is likely that the species is restricted to those two areas, where it could be restricted to an area of occupancy (AOO) of 600-800 km^2, and even if it does occur more widely in the region its AOO is not expected to exceed 2,000 km^2.

Country Occurrence:

Native, Extant (resident): Argentina

Distribution Map



Legend

EXTANT (RESIDENT)

Compiled by:

IUCN 2023

The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

The species was first recorded in 2013 in Argentina and there are only two sites in the country, Espinal forests in Córdoba province and Pampean mountains in Buenos Aires province, being considered two subpopulations. *Lysurus fossatii* from Córdoba is represented by seven collections, but the only material from Buenos Aires has been lost. It is a large species and very distinctive, and several mycological surveys have been made to the Espinal forests and to Sierra de la Ventana since its first discovery, but there are no new records of the species. It is considered that *L. fossatii* is a rare species with no clear habitat requirements. Also, the species is believed to have a restricted and disjunct distribution, occurring only in two sites within two distinct provinces in Argentina. The population size is assumed to be no more than 10,000 mature individuals, possibly being even smaller than 2,500.

Espinal forests and the Sierra de la Ventana ecosystem have been devastated by land use change by fire, agroecosystem growth for forestry, crop and cattle exploitation, and urbanization (Rodriguez-Souilla and Azaro 2021, Michalijos *et al*. 2022). Assuming a natural land cover degradation at rates of 20% in a lapse of 20 years (Guida-Johnson and Zuleta 2013), it is inferred that its population size is in continuing decline.

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

The species was found during the rainy season and described to be growing in two different phytogeographic provinces: Espinal forests, characterized by the presence of large specimens of *Prosopis alba* and *Celtis tala* and large individuals of *Morus alba*, an exotic species (Cabrera 1971; Lewis *et al*. 2004, 2009), and in Sierra de la Ventana, Buenos Aires province this area belongs to the Pampean Phytogeographic Province and it is characterized by plains and some low-lying mountains dominated by grassy species such as *Piptochaetium* and *Stipa* (Cabrera 1971).

Systems: Terrestrial

Use and Trade (see Appendix for additional information)

There is no use or trade known for this species but there has been no research about it.

Threats (see Appendix for additional information)

The Espinal Phytogeographic Province has been under human pressure including fires and deforestation for forestry, crop and cattle and urbanization (Bertonatti and Coruera 2000; Lewis *et al.* 2004, 2009; Zak 2008; Noy-Meir *et al.* 2012). In Argentina only 10% remains from the original vast extension of this dry environment (ca. 3,500 km^2) with a patchy distribution and there is only 1,100 km^2 under strict protection (Bastin *et al*. 2017, Garachana *et al*. 2018, Zeballos *et al*. 2020). Sierra de la Ventana is considered as a unique environment with high number of endemic species severely endangered by the high rates of fire events, the exotic shrub and arboreal vegetation introduction, and last decade the agroecosystem advance over this ecosystem which leads to natural land cover change and biodiversity loss (Rodriguez-Souilla and Azaro 2021, Michalijos *et al*. 2022).

Conservation Actions (see Appendix for additional information)

To prevent current population decline, the main action could be habitat preservation. Further investigation is necessary to better document the distribution and understand the biology of the species around its natural ecosystem. Moreover, the use of molecular data for phylogenetic analysis is important to test the morphology and to understand the evolutionary relationships within the genus, as well as test the viability of strains for *ex situ* conservation of the species.

Credits

Assessor(s): Hernandez Caffot, M.L., Niveiro, N., Pelissero, D., Maubet, Y., Sánchez, R., Ranieri, C., Kuhar, F. & Torres, D.

Reviewer(s): Drechsler-Santos, E. & Martins da Cunha, K.

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External Resources

For Supplementary Material, and for Images and External Links to Additional Information, please see the Red List website.

Appendix

Habitats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Habitat	Season	Suitability	Major Importance?
1. Forest -> 1.5. Forest - Subtropical/Tropical Dry	-	Suitable	-
4. Grassland -> 4.5. Grassland - Subtropical/Tropical Dry	-	Suitable	-

Plant and Fungal growth forms

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Plant and Fungal growth forms
M. Fungus

Threats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Threat	Timing	Scope	Severity
1. Residential & commercial development -> 1.1. Housing & urban areas	Ongoing	-	-
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.2. Small-holder farming	Ongoing	-	-
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
2. Agriculture & aquaculture -> 2.2. Wood & pulp plantations -> 2.2.1. Small-holder plantations	Ongoing	-	-
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
2. Agriculture & aquaculture -> 2.3. Livestock farming & ranching -> 2.3.2. Small-holder grazing, ranching or farming	Ongoing	-	-
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
7. Natural system modifications -> 7.1. Fire & fire suppression -> 7.1.1. Increase in fire frequency/intensity	Ongoing	-	-
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
7. Natural system modifications -> 7.1. Fire & fire suppression -> 7.1.3. Trend Unknown/Unrecorded	Ongoing	-	-
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		

Conservation Actions Needed

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Conservation Action Needed	Notes
1. Land/water protection -> 1.1. Site/area protection	-
1. Land/water protection -> 1.2. Resource & habitat protection	-

Research Needed

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Research Needed	Notes
1. Research -> 1.1. Taxonomy	-
1. Research -> 1.2. Population size, distribution & trends	-
1. Research -> 1.3. Life history & ecology	-
1. Research -> 1.4. Harvest, use & livelihoods	-
1. Research -> 1.6. Actions	-

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km^2): 600-2000,600-800
Number of Locations: 2
Population
Number of mature individuals: 2500-10000
Continuing decline of mature individuals: Yes
No. of subpopulations: 2
All individuals in one subpopulation: No
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Yes

The IUCN Red List Partnership

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